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Subject:Probability & Statistics

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Sec-7

Batch – 8

SUBMITTED BY:

S.No	NAME	REGD.No
1	G.Harika	241FA04281
2	Y.Bhavani	241FA04996
3	P.Gopi Krishna	241FA04B98
4	G.Pratheeka	241FA04E16

Submitted TO: DR.B.Naveen Kumar

BASIC TOPICS IN PROBABILITY:

Probability:

Definition:

Probability is a measure of how likely an event is to occur. Its value lies between 0 and 1, where 0 means impossible event and 1 means certain event.

Formula:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Example: Probability of getting head in one coin toss: $\frac{1}{2}$

Mutually Exclusive Events:

Definition

Mutually exclusive events are events that cannot happen at the same time. If one event occurs, the other event cannot occur.

Formula

If A and B are mutually exclusive:

$$P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B)$$

In the problem, “both GPUs” and “both CPUs” are mutually exclusive outcomes.

Complementary Probability:

Definition: The probability of an event happening is 1 minus the probability of it not happening.

Formula: $P(A) = 1 - P(A^c)$.

Useful when it's easier to calculate the opposite event.

Example: Probability of getting at least one CPU = $1 - P(\text{no CPUs})$. Here, “no CPUs” means both GPUs are chosen.

Sample Space and Events:

Definition:

- **Sample space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space.

Main Points to Remember:

- Every probability problem starts with identifying the sample space.
- Events can be simple (one outcome) or compound (multiple outcomes).

Example: Coin toss $\rightarrow$ Sample space = {Heads, Tails}. Event = {Heads}.

Combinations:

Definition:

A combination is a way of selecting items from a group where order does not matter.

It answers the question: “*How many different groups can I form?*”

Main Points to Remember

- Formula: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

where n = total items, r = items chosen.

- Use combinations when the arrangement/order is irrelevant.
- Compare with permutations: permutations care about order, combinations don't.

Imagine you have 4 marbles: {Red, Blue, Green, Yellow}. You want to pick 2 marbles.

Step 1: Use the formula

$$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{4 \cdot 3}{2 \cdot 1} = 6$$

So, there are 6 possible groups.

Problem Statement:

A university's computer lab has six workstations scheduled for system upgrades. Among them, two are high-performance GPUs and the remaining four are regular CPUs. Only two workstations can be upgraded on a given day, and the pair is chosen randomly from the six workstations (implying that there are 15 equally likely outcomes; for example, if the workstations are numbered 1, 2, ..., 6, then one outcome could be 1 and 2, another could be 1 and 3, and so on).

Questions:

- a) What is the probability that both selected workstations are GPU machines?
- b) What is the probability that both selected workstations are CPU machines?
- c) What is the probability that at least one selected workstation is a CPU machine?
- d) What is the probability that one GPU and one CPU workstation are selected for the upgrade?

Solution:

Given:

Total workstations = 6

GPU = 2,

CPU = 4

Number selected = 2

Step 1: Total Possible Outcomes

Since order does not matter, we use combination formula:

$${}^nC_r = \frac{n!}{r!(n-r)!} \quad {}^6C_2 = \frac{6!}{2!4!} = \frac{6 \times 5}{2 \times 1} = 15$$

Total outcomes = 15

a) Probability that both selected are GPUs

Step 1: Favorable outcomes

$$C_2 = \frac{2!}{2!0!} = 1$$

Step 2: Probability formula

$$P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}} P(\text{both GPUs}) = \frac{{}^2C_2}{{}^6C_2} = \frac{1}{15}$$

b) Probability that both selected are CPUs

Step 1: Favorable outcomes

$${}^4C_2 = \frac{4!}{2!2!} = \frac{4 \times 3}{2 \times 1} = 6$$

Step 2: Probability: $P(\text{both CPUs}) = \frac{{}^4C_2}{{}^6C_2} = \frac{6}{15}$

c) Probability that at least one selected is a CPU

Step 1: Use Complement Rule

$$P(A) = 1 - P(A')$$

At least one CPU = 1 - Probability (both GPUs)

$$P(\text{both GPUs}) = \frac{1}{15}$$

Step 2: Calculate: $P(\text{at least one CPU}) = 1 - \frac{1}{15} = \frac{14}{15}$

d) Probability that one GPU and one CPU are selected

Step 1: Favorable outcomes

Select 1 GPU:

$${}^2C_1 = 2$$

Select 1 CPU:

$${}^4C_1 = 4$$

Multiply: ${}^2C_1 \times {}^4C_1 = 2 \times 4 = 8$

Step 2: Probability $P(\text{one GPU and one CPU}) = \frac{{}^2C_1 \times {}^4C_1}{{}^6C_2} = \frac{8}{15}$

SOURCE CODE:

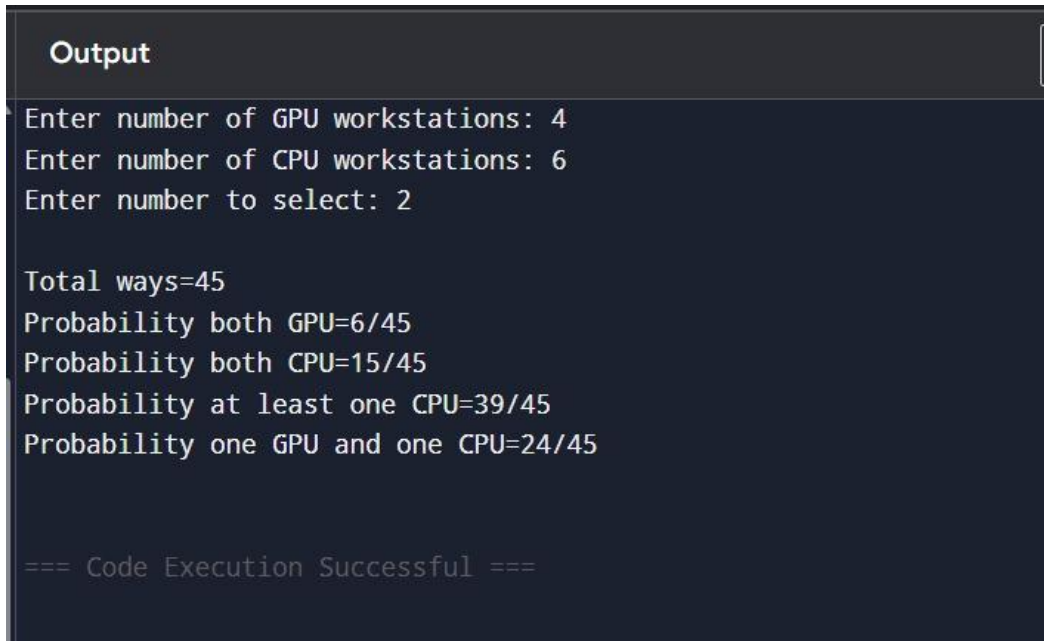
```
#include <stdio.h>

int combination(int n,int r){
int i,num=1,den=1;
for(i=1;i<=r;i++){
num=num*(n-i+1);
den=den*i;
}
return num/den;
}

int main(){
int gpu,cpu,select,total;
int totalWays,gpuWays=0,cpuWays=0,mixedWays=0;
printf("Enter number of GPU workstations: ");
scanf("%d",&gpu);
printf("Enter number of CPU workstations: ");
scanf("%d",&cpu);
printf("Enter number to select: ");
scanf("%d",&select);
total=gpu+cpu;
totalWays=combination(total,select);
if(gpu>=select){
gpuWays=combination(gpu,select);
}
if(cpu>=select){
cpuWays=combination(cpu,select);
}
if(select==2){
mixedWays=combination(gpu,1)*combination(cpu,1);
}
printf("\nTotal ways=%d\n",totalWays);
printf("Probability both GPU=%d/%d\n",gpuWays,totalWays);
```

```
printf("Probability both CPU=%d/%d\n",cpuWays,totalWays);
printf("Probability at least one CPU=%d/%d\n",totalWays-gpuWays,totalWays);
printf("Probability one GPU and one CPU=%d/%d\n",mixedWays,totalWays);
return 0;
}
```

OUTPUT:



```
Output
Enter number of GPU workstations: 4
Enter number of CPU workstations: 6
Enter number to select: 2

Total ways=45
Probability both GPU=6/45
Probability both CPU=15/45
Probability at least one CPU=39/45
Probability one GPU and one CPU=24/45

=== Code Execution Successful ===
```

CONCLUSION:

From the above calculations, the probability depends on the number of GPUs and CPUs available. Since CPUs are more than GPUs, the chance of selecting CPUs is higher. The probability of selecting both GPUs is very low, while selecting at least one CPU or both CPUs is more likely. This shows that probability is based on the number of favorable outcomes compared to the total outcomes.